AutoML for Time-Series with AutoGluon

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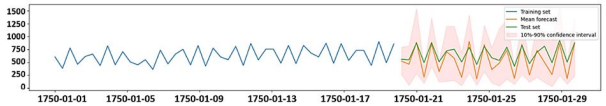
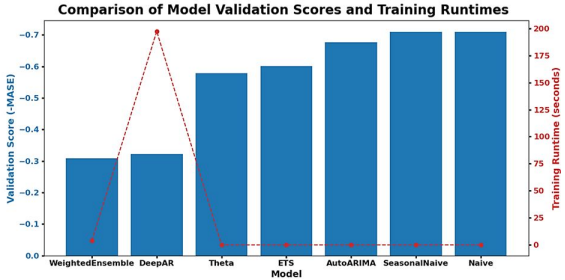
1. Abstract

- Optimize AutoML for time-series forecasting by balancing training speed with model performance
- Use AutoGluon with SMAC optimizer on the DeepAR model, employing dropout and dataset sparsity as regularization methods
- Explore model size impact on training speed and accuracy
- Apply sparsity to the dataset and observe the results



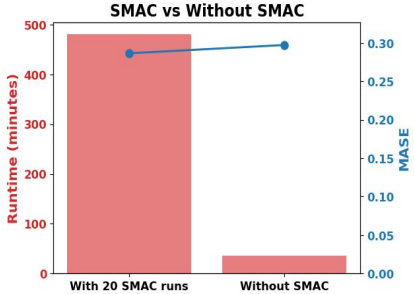
2. Which Models Perform Better?

- "m4_hourly" dataset for point and probabilistic forecasting
- Run HPO over the ML models of AutoGluon TS using SMAC
- Results does not change significantly over runtime
- Focusing on regularization, Goal to make Training faster

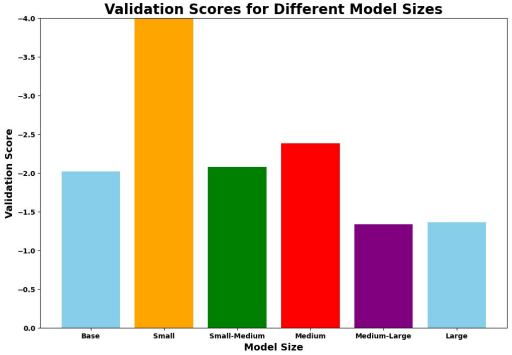


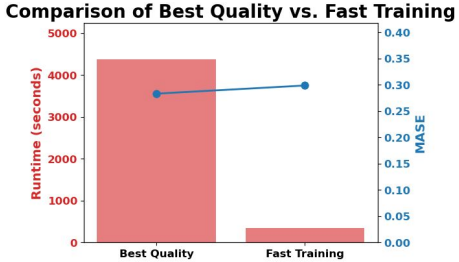
3. Does HPO Improve the Baseline Performance?

- Ran baseline on the entire dataset
- Ran SMAC with AutoGluon over DeepAR model with the following hyperparameters:
- epochs, learning rate, number of layers, number of cells, dropout_rate, scaling, context length, batch size



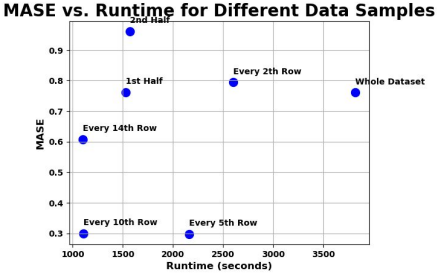
4. Does Varying Model Complexity Affect Results?





5. Does Data Sparsity Result in Better Scores?

- Applied sparsity to our dataset
- Every 5th, 10th, and 14th rows of the dataset and two halves separately



6. Conclusion

- Hyperparameter optimization showed that for this dataset, the improvement in results were not worth the time costs
- In this case, regularizing the dataset improved scores
- A potential future for this project is to test these results out across multiple datasets

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https://github.com/mehlleon/AutoML.git